MILITARY SPECIFICATION

DEVICE, ARMING-FIRING, ROCKET MOTOR

MARK 298 MOD 0

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the performance, design, development, and test requirements for the MARK 298 MOD 0 Rocket Motor Arming-Firing Device critical item, referred to herein as the A-F device.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the exact issue shown and, where not shown, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization/Documentation Division (501), Indian Head, MD 20640, by using the self-addressed Standardization Document Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 1336

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SPECIFICATIONS

MILITARY

MIL-S-6872	Soldering Process, General Specification for.
MIL-F-7179	Finishes and Coatings: Protection of Aerospace Weapons Systems, Structures and Parts; General Specification for.
MIL-B-7883	Brazing of Steels, Copper, Copper Alloys, Nickel Alloys, Aluminum and Aluminum Alloys.
MIL-E-8189	Electronic Equipment, Missiles, Boosters and Allied Vehicles, General Specification for.
MIL-W-8611	Welding, Metal Arc and Gas, Steels and Corrosion and Heat Resistant Alloys, Processes for.
MIL-I-23659	Initiators, Electric, General Design Specification for.
MIL-C-45662	Calibration System Requirements.

Naval Air Systems Command
(Code Ident 30003)

AS-5503	Reliability Program Requirements for Weapon Systems Development, General Specification for.
WS-3853A	Igniter, Rocket Motor, MARK 264 MOD 1.
WS-6536	Procedures and Requirements for Preparation and Soldering of Electrical Connections.

Naval Sea Systems Command
(Code Ident 53711)

WS-18846	Squib, Eelectric, Dual Bridgewire, MARK 18 MOD 0.

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OD-30393	Design Principles and Practices for Controlling Hazards of Electromagnetic Radiation to Ordnance (HERO Design Guide).

STANDARDS

FEDERAL

FED-STD-H28	Screw-Thread Standards for Federal Services.
FED-STD-102	Preservation, Packaging, and Packing Levels.
FED-STD-595	Federal Standard Colors.

MILITARY

MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-130	Identification Marking of U.S. Military Property.
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of.
MIL-STD-167-1	Mechanical Vibrations of Shipboard Equipment (Type I-Environmental and Type II-Internally Excited).
MIL-STD-331	Fuze and Fuze Components, Environmental and Performance Tests for.
MIL-STD-453	Inspection, Radiographic.
MIL-STD-454	Standard General Requirements for Electronic Equipment.
MIL-STD-785	Reliability Program for Systems and Equipment Development and Production.

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MIL-STD-810	Environmental Test Methods.
MIL-STD-889	Dissimilar Metals.
MIL-STD-1695	Environments, Working, Minimum Standards for.

DRAWINGS

Naval Air Systems Command
(Code Ident 30003)

DL 1568376	Assembly, Igniter.
DL 3300561	Rocket Motor, Arming-Firing Device, MARK 298 MOD 0.
DL 642AS6053	Assembly, Solenoid.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 171-178	Transportation.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

3. REQUIREMENTS

3.1 Item definition. The A-F device remotely arms and initiates a rocket motor, and consists of a housing assembly and an igniter assembly, defined as follows:

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(a) Housing assembly. The housing assembly contains and supports the components of the A-F device, serves as the forward closure for the rocket motor, and provides the interface between the rocket motor and the igniter assembly. Provisions are included in the housing assembly for environmental sealing of the rocket motor and A-F device components, observing the visual indicator, and electrically connecting the A-F device to the launch cannister. The housing assembly contains an arming-firing mechanism consisting of the following components:

(1) Solenoid assembly. The solenoid assembly (DL 642AS6053), with return spring and housing, rotates the switch and rotor upon receiving the arm command.

(2) Rotor. The rotor is the rotating portion of the mechanical barrier and is mounted on the solenoid shaft.

(3) Switch. The switch is actuated by the solenoid assembly and electrically connects the squib to the firing circuit.

(4) Initiator assembly. The initiator assembly consists of the MARK 18 MOD 0 Squib which is in accordance with WS-18846 and qualified to the requirements of MIL-I-23659C, Class A, a welded-on pressure sleeve which provides the required free volume for fail-safe pressure containment, the seals, the nonrotating portion of the mechanical barrier, an electrical conductor for two parallel firing circuits, and a poppet valve.

(b) Igniter assembly. The igniter assembly mounts to the A-F device housing assembly and contains the igniter pyrotechnics which ignite the rocket motor grain. The igniter assembly utilizes the standard pyrotechnic components from the MARK 264 MOD 1 Igniter (booster charge and igniter grain) which is initiated by the MARK 18 MOD 0 Squib contained in the initiator assembly. The use of this proven pyrotechnic train will result in reliable rocket motor ignition.

3.2 Characteristics.

3.2.1 Performance.

3.2.1.1 Safe condition (see 6.4.1). The A-F device shall remain in the safe condition during all Rolling Airframe Missile (RAM) environmental conditions, unless the arm command is present.

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3.2.1.1.1 Mechanical seal. When the A-F device is intentionally modified and fired in the safe condition, the A-F device housing assembly shall not exhibit a leak rate greater than 1×10^{-4} standard atmospheric cubic centimeters per second (std atm cm^3/s).

3.2.1.1.2 Electrical. While in the safe condition, the fire and return lines to each squib bridgewire shall be shorted together, which will eliminate spurious electrical signals from initiating the squib. While in the safe condition, the monitor circuit shall present a continuous electrical path.

3.2.1.2 Armed condition (see 6.4.2). Arming shall be initiated by the arm command which shall energize the solenoid. The solenoid shall then rotate the mechanical barrier so the barrier pins are aligned with the holes in the mechanical barrier.

3.2.1.2.1 Mechanical, armed condition. The A-F device shall arm within 100 milliseconds (ms) maximum from the arm command, and shall be considered armed whenever the initiator assembly is capable of translating past the rotor portion of the mechanical barrier. Upon removal of the arm command, the A-F device shall return to the safe condition.

3.2.1.2.2 Electrical, armed condition. The rotary motion of the solenoid shall simultaneously operate the switch. The switch shall complete the firing circuit to the squib after, or at the same time as, the A-F device arms mechanically.

3.2.1.3 Firing. The A-F device shall have a separate firing circuit that is isolated from the missile system ground. The electrical circuit shall be capable of supplying 5 amperes (A) minimum to each of the squib bridgewires.

3.2.1.3.1 Squib. Upon squib initiation in the armed condition, the initiator assembly shall translate forward and open the poppet valve, allowing the squib output gases to reach the igniter booster charge.

3.2.1.3.2 Igniter booster. The squib shall initiate the igniter booster pyrotechnic material which shall produce a pressure of 500 pounds per square inch (lb/in^2) minimum when fired into a fixture that simulates the igniter tube.

3.2.1.3.3 Igniter grain. The igniter booster charge shall initiate the igniter grain. The output of the A-F device shall produce an output equivalent to the MARK 264 MOD 1 Igniter when fired into the test fixture specified in WS-3853A.

3.2.1.4 Life expectancy. The A-F device shall be designed to have a life expectancy of 5 years minimum.

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3.2.1.5 Radiographic. The A-F device shall show no evidence of cracks, voids or fissures, and shall meet the workmanship requirements specified herein.

3.2.2 Physical characteristics. The design concept of the A-F device is shown on Drawing 3300561.

3.2.2.1 Dimensions and weight. The dimensions and weight of the A-F device shall be as follows:

(a) Length: 13 inches maximum.

(b) Diameter: 4.385 inches maximum at motor closure.

(c) Weight: To be determined (TBD).

3.2.2.2 Mechanical. The mechanical design shall be as follows.

3.2.2.2.1 Initiator assembly. The initiator assembly shall include a free volume so that squib pressure is limited to a safe value in the event of firing in the safe condition. The initiator assembly shall be secured to maintain the proper angular relationship between the rotating and nonrotating components of the mechanical barrier. The initiator assembly shall be secured to prevent axial translation before squib firing.

3.2.2.2.2 Mechanical barrier. The mechanical barrier shall withstand a squib generated pressure of 10,000 lb/in^2 when the A-F device is intentionally modified and fired in the safe condition. The resulting deformations shall not allow the initiator assembly to move more than 0.050 inch. The rotor portion of the mechanical barrier shall be keyed and secured to the solenoid assembly to ensure proper alignment.

3.2.2.2.3 Solenoid assembly. The solenoid assembly (DL 642AS6053) shall meet the following requirements:

(a)	Rotation:	45 ±3 degrees at applied voltage.
(b)	Net torque:	1.0 ounce-inch (oz-in) minimum at minimum applied voltage.
(c)	Return-spring torque:	1.5 oz-in minimum.

3.2.2.2.4 Housing assembly. The housing assembly shall withstand, without permanent deformation, an internal pressure of 10,000 lb/in^2 resulting from squib initiation. The housing assembly shall maintain structural integrity throughout the flight of the missile. The housing assembly and the igniter tube shall not touch the motor grain or inhibitor coating.

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3.2.2.2.5 Seal. The A-F device housing-to-motor-tube interface (seal) shall be in accordance with Drawing SK 3300570. The A-F device internal cavity, containing the solenoid assembly, switch, and mechanical barrier, shall be protected by environmental seals with a leak rate which shall not exceed 1×10^{-4} std atm cm^3/s.

3.2.2.2.6 Igniter assembly. The igniter assembly shall be manufactured using the pyrotechnic components specified on DL 1568376.

3.2.2.3 Electrical. The electrical characteristics of the A-F device shall be in accordance with Drawing 3300562 and as follows.

3.2.2.3.1 Electrical connector. The A-F device electrical connector shall be designed to be compatible with the launch cannister electrical connector and circuitry. The design shall not require any wire connections to be made after the A-F device is installed in the rocket motor, and shall provide electromagnetic shielding, as specified in 3.3.2.

3.2.2.3.2 Electrical interface characteristics. The A-F device electrical interface characteristics shall be as follows:

(a) Arming circuit:

(1) 28.0 ±2.5 ohms.

(2) Solenoid case insulation resistance: 100 megohms (Mohms) minimum at 500 volts direct current (Vdc).

(3) Input voltage: 26 ±4 Vdc.

(4) Minimum resultant current handling capability: 1.2 A continuous.

(5) Back-to-back zener diode for inductive kick suppression.

(b) Firing circuit:

(1) Resistance: 0.65 ±0.20 ohm.

(2) Input voltage: 26 ±4 Vdc.

(3) Current handling capability: 10 A minimum.

(c) Safe and armed position monitoring circuit:

(1) Resistance: 0.5 ohm maximum.

(2) Current handling capability: 0.5 A continuous.

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3.2.3 Reliability. The A-F device shall be designed to meet the performance requirements of 3.2.1 following the environmental conditions of 3.2.5, with a reliability goal of 0.99 and a confidence level of 95 percent. Unless otherwise specified in the contract or purchase order, performance data resulting from testing of prototype units, preproduction samples, and production lot samples shall be used for reliability determination (see 6.2.1). Reliability oriented design techniques, such as stress derating, redundancy, stress-strength analysis, and tolerance analysis, shall be used with emphasis on critical components or potentially critical failure modes.

3.2.3.1 Reliability program plan. A reliability program plan in accordance with AS-5503 and MIL-STD-785 shall be conducted to ensure that the A-F device meets the reliability requirements specified herein (see 6.2.2).

3.2.4 Maintainability. The A-F device shall be designed for zero maintenance during the life cycle.

3.2.5 Environmental conditions. The A-F device shall not be damaged, nor shall safety or subsequent performance be degraded, by exposure to any natural combination of service use environments. These shall include, but not be limited to, the following environments.

3.2.5.1 Transportation vibration - temperature. The A-F device shall meet the requirements of MIL-STD-331, Test 119, Procedure 1, and shall be safe and operable (see 6.4.3) after the test.

3.2.5.2 Shipboard vibration. The A-F device shall meet the requirements of MIL-STD-167-1, Type I, and shall be safe and operable after the test.

3.2.5.3 Temperature and humidity. The A-F device shall meet the requirements of MIL-STD-331, Test 105.1, either method, and shall be safe and operable after the test.

3.2.5.4 Shock. The A-F device shall meet the requirements of MIL-STD-810, Method 516.2, Procedure IV, Figure 516.2-1 (amplitude a, time duration c), without performance degradation or malfunction during and after the test.

3.2.6 Transportability. The A-F device shall be designed to be transported in a packaged condition by any appropriate transportation method available for pyrotechnic materials and in accordance with 49 CFR 171-178.

3.3 Design and construction. The A-F device shall be designed to minimize complexity and to ensure maximum safety, reliability, efficiency, ease of operation, handling, and economical production. The drawings and documents listed on DL 3300561 and specified on Drawing 3300561 define and describe one method of manufacturing the A-F device.

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3.3.1 Materials, processes, and parts.

3.3.1.1 Moisture and fungus resistance. Materials shall be resistant to the corrosive action of salt air and damage from moisture and fungus in accordance with MIL-E-8189. Protective coatings shall be acceptable as moisture, fungus, and corrosion preventatives on parts which will not lose a coating during the normal course of examination, operation, and periodic test.

3.3.1.2 Materials. Materials shall be in accordance with existing Government specifications wherever possible. For materials not covered by Government specifications, the requirements shall be as detailed on the applicable drawings.

3.3.1.3 Dissimilar metals. The A-F device shall be designed for maximum corrosion resistance between dissimilar metals in accordance with MIL-STD-889, and shall be insulated in accordance with MIL-F-7179 wherever contact is unavoidable.

3.3.1.4 Castings. Castings shall not be used for parts which are stressed more than 50 percent of the material yield strength. The applicable drawings specify where castings shall be used.

3.3.1.5 Plastics. The use of plastics within the A-F device shall be limited to those materials which will not outgas corrosively, creep under the applied load, nor crack due to aging.

3.3.1.6 Threads. Threads and the go/no-go gaging practice shall be in accordance with FED-STD-H28. The class or fit for threads shall be as specified on the drawings.

3.3.1.7 Soldering. Soldering of all electrical connections shall be in accordance with WS-6536. All mechanical soldering shall be in accordance with MIL-S-6872.

3.3.1.8 Wiring. Wiring and terminal design shall include protection against environmental and mechanical hazards to prevent corrosion or shorting of circuits. Electrical connectors shall be capable of being connected and disconnected and the cables flexed without damage within the storage temperature range from -80 to +167 degrees Fahrenheit (°F).

3.3.2 Electromagnetic radiation. Using OD-30393 as a guide, accepted design practices shall be followed when designing the A-F device to reduce susceptibility to electromagnetic radiation.

3.3.3 Nameplates and product marking. Nameplates and product marking shall be in accordance with MIL-STD-130.

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3.3.4 Workmanship. The A-F device shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, marking of parts, wiring, welding and brazing, plating, riveting, finishes, machine-screw assemblage, freedom from burrs, sharp edges, dirt, chips, or other foreign matter and visible defects which are likely to affect or render the A-F device unsatisfactory and nonserviceable over the useful life for the operation or function intended. Workmanship for electronic equipment shall be in accordance with MIL-STD-454, Requirement 9.

3.3.4.1 Screws. Screws shall be the self-locking type, wherever possible.

3.3.4.2 Welds. Welds shall be in accordance with MIL-W-8611.

3.3.4.3 Brazing. Brazing shall be in accordance with MIL-B-7883.

3.3.4.4 Cleanup. Prior to and after final assembly, all subassembly parts and the assembled A-F device shall be thoroughly cleaned of any foreign matter.

3.3.4.5 Facilities and working environment. The contractor shall provide adequate facilities for the fabrication, assembly, testing and examination of the A-F device. Unless otherwise specified in the contract or purchase order, as a minimum, the working environment shall meet the levels and requirements specified in MIL-STD-1695 (see 6.2.1).

3.3.5 Interchangeability. Mechanical and electrical interchangeability shall exist between similar assemblies or subassemblies of the A-F device. Interchangeability requires the substitution of similar parts without physical modification to any part of the component, including cabling, wiring, and mounting.

3.3.6 Safety. The safety requirements for the A-F device shall be as follows.

3.3.6.1 Remotely-operated safety mechanism. The A-F device shall incorporate a remotely-operated safety mechanism which provides for the following conditions in the ignition system of a rocket motor:

(a) Safe. The squib shall be mechanically isolated from the igniter booster charge, and electrically isolated from the firing circuit.

(b) Armed. The A-F device pyrotechnic train shall be aligned prior to, or at the same time as, completion of the electrical circuit to the squib. The electrical path between the firing supply and the squib shall be continuous.

3.3.6.2 Auto-return to safe. The A-F device solenoid shall be designed to automatically return to the safe condition from the armed condition upon removal of the arm command. The solenoid spring shall be included in the A-F device to prevent accidental movement of the mechanism from one position to another position.

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3.3.6.3 Shorted firing circuit. The design shall include an electrically shorted squib bridgewire circuit that remains in the shorted position while the A-F device is in the safe condition.

3.3.6.4 Visual indicator. The A-F device shall have a visual indicator that allows the position of the mechanical barrier to be determined when assembled in the all-up round configuration (missile inside the launch cannister). The safe position shall be indicated by a white (FED-STD-595, Color 27875) letter S on a green (FED-STD-595, Color 38901) background. The armed position shall be indicated by a white (FED-STD-595, Color 27875) letter A on a red (FED-STD-595, Color 38903) background. The A-F device shall be designed so the visual indicator markings are readily discernible from outside the rocket motor to maximize readability and minimize errors.

3.3.6.5 Safety tests.

3.3.6.5.1 Drop test, 1.5 meter (five-foot drop). The A-F device shall meet the requirements of MIL-STD-331, Test 111.1, Procedure 1, Criterion B, and shall be safe and operable after the test.

3.3.6.5.2 Forty-foot drop. The A-F device shall meet the requirements of MIL-STD-331, Test 103.1, Procedure 1, and shall be safe to handle and dispose of after the test.

3.3.6.5.3 Jolt. The A-F device shall meet the requirements of MIL-STD-331, Test 101.2, Procedure II, and shall be safe to handle, store, and transport after the test.

3.3.6.5.4 Auto-ignition. The A-F device pyrotechnics shall not ignite while the A-F device is exposed to a temperature of 350 +10, -0°F for 2 hours minimum.

3.3.7 Human performance/human engineering. The A-F device shall be designed so the safe or armed condition can be verified easily by a visual examination of the rocket motor, as specified in 3.3.6.4.

3.4 Documentation.

3.4.1 Design approval drawings. The design approval drawings shall include sufficient information to proceed with design and development, and shall be superseded by production drawings at the beginning of A-F device production. When specified in the contract or purchase order, the requirements for documenting the design shall be as specified on the applicable Contract Data Requirements List, DD Form 1423 (see 6.2.2).

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3.5 Precedence.

3.5.1 Documentation. Specifications, standards, and drawings shall be used in the order of precedence in accordance with MIL-STD-143. When the requirements of the contract, this specification, or applicable subsidiary documents are in conflict, the following precedence shall apply:

(a) Contract. The contract shall have precedence over any other document.

(b) This specification. This specification shall have precedence over all applicable subsidiary documents.

(c) Referenced documents. Any document referenced in this specification shall have precedence over all applicable subsidiary documents referenced therein.

4. QUALITY ASSURANCE PROVISIONS

4.1 General. This section specifies the requirements for verification of the performance and design characteristics of the A-F device. The final design and all elements of the design shall be subject to the approval of the procuring activity. The development hardware shall be subjected to environmental and ignition testing to prove the adequacy of the design. Appropriate changes shall be made to correct any deficiencies, and testing shall be conducted to verify the adequacy of any changes.

4.1.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure that supplies and services conform to prescribed requirements (see 6.2.1).

4.1.1.1 Responsibility for engineering model tests. Unless otherwise specified in the contract or purchase order, the engineering model tests of Figure 1 will be performed by the procuring activity (see 6.2.1).

4.1.1.2 Responsibility for pyrotechnic train verification tests. Unless otherwise specified in the contract or purchase order, the pyrotechnic train verification tests of Figure 2 shall be performed by the contractor (see 6.2.1).

4.1.1.3 Responsibility for closed bomb ignition tests. Unless otherwise specified in the contract or purchase order, the closed bomb ignition tests shall be performed by the contractor (see 6.2.1).

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4.1.2 Special tests and examinations. During development, tests and examinations shall be performed to verify that the requirements of Section 3 have been met. When specified in the contract or purchase order, detailed test plans and test procedures for the pyrotechnic train verification tests and closed bomb ignition tests shall be prepared by the contractor and shall be approved by the procuring activity prior to use (see 6.2.2).

4.1.2.1 Engineering model tests. Five A-F devices shall be subjected to the tests specified in Figure 1 to verify that the requirements of 3.2.1, 3.2.2, 3.2.5, and 3.3.6 have been met.

4.1.2.2 Pyrotechnic train verification tests. Twenty-four pyrotechnic train units shall be tested in accordance with a test plan and test procedure approved by the procuring activity (see 4.1.2 and 6.2.2) to verify that the pyrotechnic train meets the requirements of 3.2.1.3.1 and 3.2.1.3.2. For the test sequence of Figure 2, overload is defined as using a MARK 18 MOD O Squib in which the output charge has been increased by 30 percent over the nominal requirement of 195 milligrams (mg). Underload is defined as using a MARK 18 MOD O Squib in which the output charge has been reduced by 30 percent under the nominal requirement of 195 mg.

4.1.2.3 Closed bomb ignition tests. The purpose of these tests is to establish baseline performance data for the A-F device that can be compared to existing MARK 264 MOD 1 Igniter test results. The test shall consist of at least 5 A-F devices fired in the test fixture specified in WS-3853A, so the parameters of WS-3853A are obtained. The A-F device fixture shall simulate the pyrotechnic train and the configuration of the actual A-F device.

4.1.3 Test provisions and conditions. When specified in the contract or purchase order, all test equipment shall be calibrated and maintained in accordance with a plan meeting the requirements of MIL-C-45662. The calibration plan shall be approved by the procuring activity prior to use (see 6.2.2).

4.1.3.1 Test facility environment. Unless otherwise specified herein, all tests shall be performed under the following environmental conditions:

(a)	Temperature:	Ambient (between 65 and 85°F).
(b)	Altitude:	Elevation of test facility.
(c)	Relative humidity:	Maintained between 10 and 70 percent.

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4.1.3.2 Test equipment accuracy. Test equipment and inspection facilities shall be of sufficient accuracy, quality, and quantity to permit performance of the required inspections. Unless otherwise specified in the contract or purchase order, the limit of error in the measuring apparatus shall not exceed 10 percent of the tolerance specified for the parameter being measured (see 6.2.1).

4.2 Quality conformance inspections. The following tests and examinations shall be performed to determine that all of the requirements of Section 3 have been met.

4.2.1 Quality tests and examinations.

4.2.1.1 Visual examination. Each A-F device shall be visually examined by the contractor for conformance to workmanship and the general requirements of Section 3.

4.2.1.2 Seal test and inspection. Each A-F device shall be inspected and tested by the contractor in accordance with MIL-STD-331, Test 118, Bubble Method. To be acceptable, the seal shall meet the requirements of 3.2.1.1.1 and 3.2.2.2.5.

4.2.1.3 Radiographic inspection. Each A-F device will be radiographically inspected by the procuring activity in accordance with MIL-STD-453 where specified on Figure 1. To be acceptable, the A-F device shall meet the requirements of 3.2.1.5.

4.2.2 Environmental tests. Environmental tests will be performed by the procuring activity where specified on Figure 1 to verify the adequacy of the design.

4.2.2.1 Transportation vibration - temperature. The A-F device shall be tested in accordance with MIL-STD-331, Test 119, Procedure 1. To be acceptable, the A-F device shall meet the requirements of 3.2.5.1.

4.2.2.2 Shipboard vibration. The A-F device shall be tested in accordance with MIL-STD-167-1, Type I. To be acceptable, the A-F device shall meet the requirements of 3.2.5.2.

4.2.2.3 Temperature and humidity. The A-F device shall be tested in accordance with MIL-STD-331, Test 105.1, either method. To be acceptable, the A-F device shall meet the requirements of 3.2.5.3.

4.2.2.4 Shock. The nonoperating A-F device shall be tested in accordance with MIL-STD-810, Method 516.2, Procedure IV, Figure 516.2-1 (amplitude a, time duration c). To be acceptable, the A-F device shall meet the requirements of 3.2.5.4.

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4.2.3 Safety tests. Safety tests will be performed by the procuring activity where specified on Figure 1 to verify the adequacy of the design.

4.2.3.1 Drop test, 1.5 meter (five-foot drop). The A-F device shall be mounted in an inert test vehicle and tested in accordance with MIL-STD-331, Test 111.1, Procedure 1, Criteron B. To be acceptable, the A-F device shall meet the requirements of 3.3.6.5.1.

4.2.3.2 Forty-foot drop. Two A-F devices shall be mounted in an inert test vehicle and tested in accordance with MIL-STD-331, Test 103.1, Procedure 1. One A-F device shall be dropped with the forward end down and one A-F device shall be dropped horizontally. To be acceptable, the A-F device shall meet the requirements of 3.3.6.5.2.

4.2.3.3 Jolt. The A-F device shall be tested in accordance with MIL-STD-331, Test 101.2, Procedure II. To be acceptable, the A-F device shall meet the requirements of 3.3.6.5.3.

4.2.3.4 Visual indicator. The A-F device shall be tested by mounting in an inert test vehicle and observing the visual indicator. To be acceptable, the A-F device shall meet the requirements of 3.3.6.4 and 3.3.7.

4.2.3.5 Auto-ignition. The A-F device shall be tested where specified on Figure 1. To be acceptable, the A-F device shall meet the requirements of 3.3.6.5.4.

4.2.3.6 Mechanical barrier. The A-F device shall be fired in the safe condition where shown on Figure 1. To be acceptable, the A-F device shall meet the requirements of 3.2.2.2.2 and 3.2.2.2.4.

4.2.4 Performance/function tests (see 6.3).

4.2.4.1 Electrical. Each A-F device shall be tested by the contractor to determine conformance with 3.2.1.1.2 and 3.2.2.3.

4.2.4.2 Arming. Each A-F device shall be tested by the contractor to determine conformance with 3.2.1.2

4.2.4.3 Firing. The A-F device will be tested for pyrotechnic output by the procuring activity where specified on Figure 1. To be acceptable, the A-F device shall meet the requirements of 3.2.1.3.

4.3 Packaging, packing, and marking. Prior to shipment, examination shall be made to ensure that packaging, packing, and marking conform to the requirements of Section 5.

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5. PACKAGING

5.1 Preservation packaging. Preservation and packaging shall be in accordance with FED-STD-102.

5.1.1 Level C. The A-F device housing assemblies and igniter assemblies shall be clean, dry, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing.

5.2.1 Level C.

5.2.1.1 Housing assemblies. The A-F device housing assemblies, packaged in accordance with 5.1.1, shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. Unless otherwise specified in the contract or purchase order, the quantity of housing assemblies in each container shall be at the option of the contractor (see 6.2.1). Packing shall conform to 49 CFR 171-178.

5.2.1.2 Igniter assemblies. The A-F device igniter assemblies, packaged in accordance with 5.1.1, shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. Unless otherwise specified in the contract or purchase order, the quantity of igniter assemblies in each container shall be at the option of the contractor (see 6.2.1). Packing shall conform to 49 CFR 171-178.

5.3 Marking. In addition to any special marking required by the contract or purchase order, all marking shall be in accordance with MIL-STD-129 (see 6.2.1).

6. NOTES

6.1 Intended use. The A-F device is intended for use in the XRIM-116A Rolling Airframe Missile (RAM) rocket motor of the EX-31 Guided Missile Weapon System.

6.2 Ordering data.

6.2.1 Procurement requirements. Procurement documents should specify the following:

(a) Number, title, and date of this specification.

(b) Reliability, if other than as specified in 3.2.3.

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(c) Working environment, if other than as specified in 3.3.4.5.

(d) Responsibility for inspection and inspection facilities, if other than as specified in 4.1.1.

(e) Responsibility for engineering model tests, if other than as specified in 4.1.1.1.

(f) Responsibility for pyrotechnic train verification tests, if other than as specified in 4.1.1.2.

(g) Responsibility for closed bomb ignition tests, if other than as specified in 4.1.1.3.

(h) Test equipment accuracy, if other than as specified in 4.1.3.2.

(i) Quantity of housing assemblies and igniter assemblies in each container (see 5.2.1.1 and 5.2.1.2).

(j) Special marking when required (see 5.3)

6.2.2 Data requirements. When this specification is used in a procurement which incorporates a Contract Data Requirements List (DD Form 1423) and invokes the provisions of 7-104.9(n) of the Defense Acquisition Regulations (DAR), the data requirements identified below will be developed as specified by an approved Data Item Description (DID) (DD Form 1664) and delivered in accordance with the approved DD Form 1423 incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited as follows:

Paragraph	Data Requirement	Applicable DID
3.2.3.1	Reliability Program Plan	UDI-R-2113
3.4.1	Design Approval Drawings	DI-E-7031
4.1.2	Test Plan and Test Procedures	DI-T-5204
4.1.3	Calibration Plan	UDI-T-23742A

(Copies of DIDs required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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6.3 Explosives safety precautions. Minimum explosives safety precautions for use by the contractor are detailed in DoD Instruction 4145.26M, DoD Contractors' Safety Manual for Ammunition, Explosives and Related Dangerous Material.

6.4 Definitions.

6.4.1 Safe condition. The A-F device is considered to be in the safe condition when the holes in the mechanical barrier are out-of-line with the barrier pins, and when the squib bridgewires are disconnected from the firing lines and shorted together.

6.4.2 Armed condition. The A-F device is considered to be in the armed condition when the holes in the mechanical barrier are in line with the barrier pins, and the firing circuit is connected to the squib bridgewires.

6.4.3 Operable. The A-F device is considered to be operable when the solenoid rotates to the armed condition within 100 ms after applying an arm command of 22 +0, -1 Vdc, and then returns to the safe condition within 50 ms after removal of the arm command.

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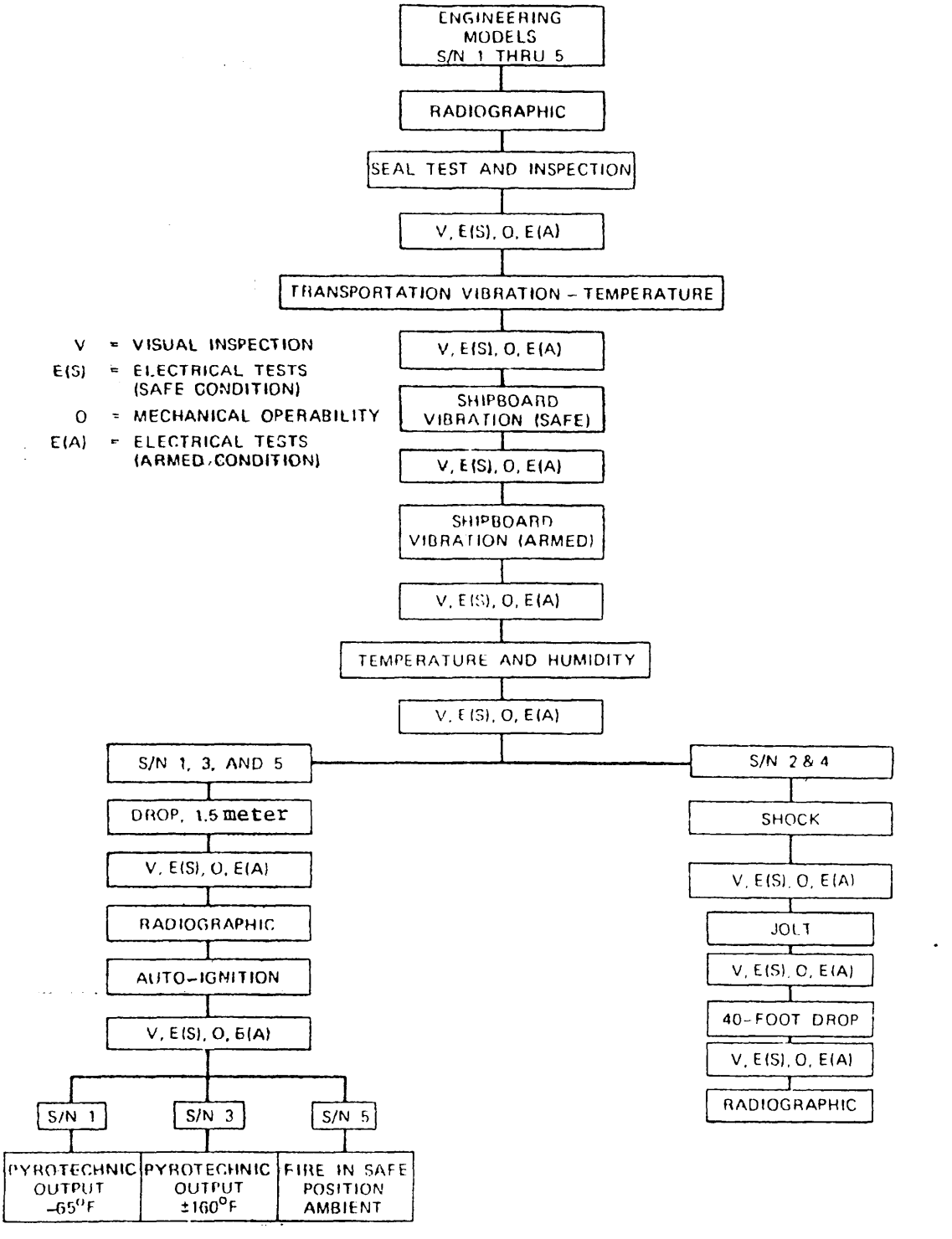


FIGURE 1. A-F device engineering model test sequence.

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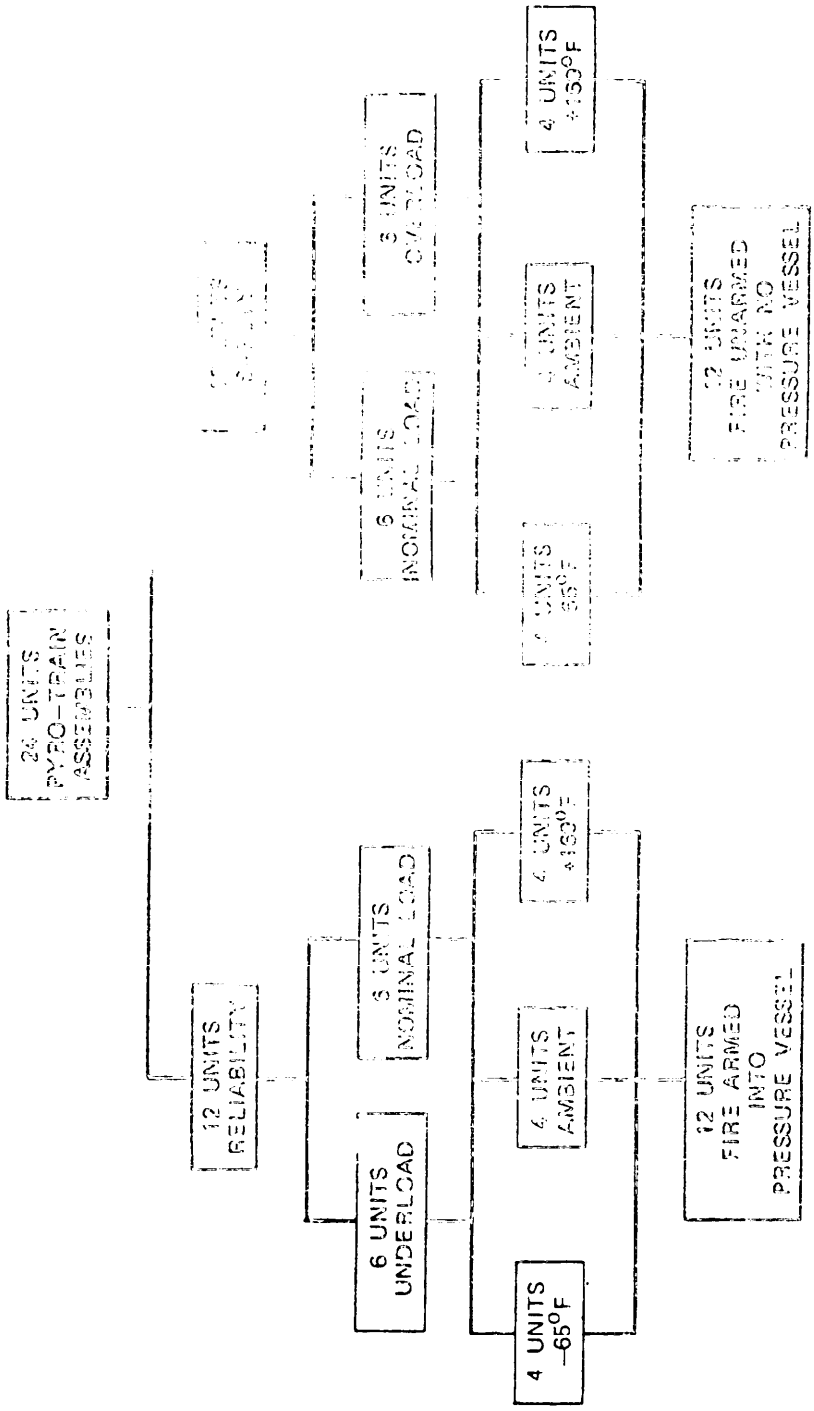


FIGURE 2. Pyrotechnic train verification test sequence.

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